

Deep Learning for Short-Term Electricity Peak Demand Prediction

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1 Introduction

Electric systems are increasingly challenged growing reliance of renewable energy sources and therefore weather variability, and population growth. These all contribute to uncertain load patterns. In this context, accurately predicting short-term electricity demand is critical for grid stability, and importantly, reducing the risk of overloads. Determining future demand is a highly decision-relevant task for utility operators and energy markets.

Deep learning is uniquely suited for this task because electricity consumption exhibit highly non-linear and stochastic behaviors. Traditional statistical methods often struggle to simultaneously capture weekly patterns (weekdays vs. weekends) and complex weather interactions (e.g., compounding heatwaves). A Gated Recurrent Unit (GRU), is chosen due to its ability to adapt dynamically to energy output differences, outperforming traditional models [1]. A GRU excels at sequence-to-sequence regression since it can maintain an internal hidden state that captures the underlying dynamics.

The objective of this project is to predict the daily electricity demand profile for the upcoming week (a 7-day forecast), the GitHub pipeline found in [2]. This will use a multivariate time series of historical load data, calendar contexts, and weather conditions within Ontario.

2 Background & Related Work

Broadly, deep learning techniques have been applied to energy optimization [3]. A couple examples of this are bellow.

- A hybrid short-term load forecasting framework that combines deep learning with heuristic optimization [4]. Model parameters were optimized using a genetic wind-driven algorithm, achieving improved short-term forecasting accuracy.
- At the household level, hybrid machine learning approaches have been applied to forecast appliance-level electricity consumption [5].

- Hybrid data-mining approaches [6] have been used to forecast yearly peak demand.
- Prior work during deep learning for peak forecasting [7, 8] that helped inspire this project.

3 Data Processing

Data Sources & Collection Challenges:

The primary dataset for this project consists of three merged sources spanning from 2002 to 2025. Historical Ontario electricity demand was sourced from Kaggle [9] (originally at an hourly resolution). For weather features, the initial attempt was to automate download hourly data from the Environment and Climate Change Canada historical archive [10]. However, automated requests were intercepted by server security measures, causing tokenization failures. Therefore, a pivot to *toronto.weatherstats.ca* [11] still allowed reasonable weather data. Additionally, monthly population data was used to account for long-term grid load expansion from [12].

Cleaning and Formatting: To align the datasets, the hourly electricity demand was aggregated into daily averages. The weather dataset contained several non-numeric values, such as 'M' for missing, which were coerced into NaN values. The population growth data was provided as strings containing percentage symbols, cast to floats. A forward-fill mechanism was implemented to maintain continuity using zeros. To align the monthly population data, the monthly values were broadcasted across all respective days. A 'month' feature was also added to assist the model's ability to capture seasonality. Since data was transformed from hourly to daily, the objective shifted from predicting daily demand (as explained in the Project Proposal) to weekly demand.

Splitting Strategy: Unlike image datasets, time-series data is highly susceptible to temporal data leakage. Therefore, a randomized split was avoided. The data was partitioned chronologically:

- **Training Set (2002–2018):** Used for model fitting.
- **Validation Set (2019–2021):** Used for

hyperparameter tuning.

- **Test Set (2022–2025):** The "never-before-seen" data.

Normalization & Tensor Generation:

To ensure stable gradient descent, all 15 features were scaled to a $[0, 1]$ range using `MinMaxScaler`. Crucially, the scaler was fitted only on the training set and subsequently applied to the validation and test sets to prevent future data leakage. Lastly, the data was transformed into 3D PyTorch tensors. The model receives a 14-day historical sequence (15 features per day).

Dataset Split	Days	Tensor Shape (X)
Training (2002–2018)	6,089	$[6068, 14, 12]$
Validation (2019–2021)	1,096	$[1075, 14, 12]$
Test (2022–2025)	1,461	$[1440, 14, 12]$

Figure 1: Chronological data split statistics and resulting 3D sequence tensor dimensions $[Batch, Seq, Feat]$

An example of a cleaned training sample sequence consists of a 14-day 14×15 matrix of normalized past demand, population, and weather context, paired with a target y vector of shape [7] representing the exact daily demands.

4 Primary Model

The primary architecture is a Gated Recurrent Unit (GRU) network, selected for its efficiency in modeling non-linear temporal dependencies.

Model Inputs and Outputs: The system is designed to take a localized window:

- **Input (X):** A 14-day historical sequence. Each day is represented by a 12-feature vector ($X_t \in R^{12}$), containing daily average electricity demand, temporal data (month, year, day of week, and weekend indicator), socio-economic context (population growth), and meteorological variables (average temperature, relative humidity, wind speed, health index, rain, and snow).
- **Output (y):** A vector of 7 continuous scalar values representing the predicted average electricity demand for each day of the following week.

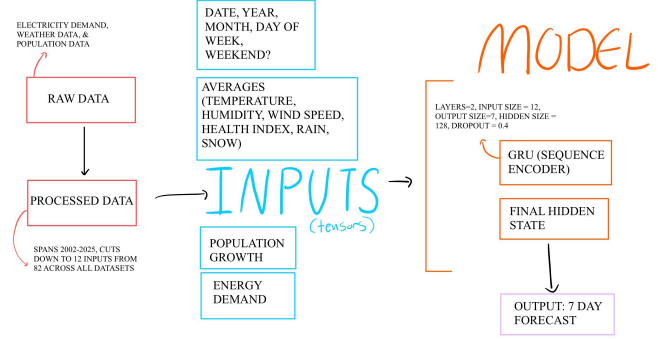


Figure 2: General overview of the project pipeline and architecture.

Although such models should be used as decision-support tools rather than autonomous grid controllers [13], this sequence-based framework provides a data-driven approach to short-term demand forecasting without relying on heavily engineered hybrid pipelines.

4.1 Reproducibility

The model utilizes a stacked GRU structure to process a 14-day look-back window.

- **Input Shape:** 12 features, listed in Section 1.
- **Regularization:** A Dropout rate of 0.4 is applied between GRU layers. Total trainable parameters are approximately 110,000.
- **Training:** Optimized via the Adam algorithm ($\eta = 0.001$) using MSE loss. All features were normalized via `MinMaxScaler` fit only on the training set (2002–2018).

4.2 Hyperparameters

A grid search was performed over hidden sizes and learning rates. As displayed in the Figure 3, the 128-unit architecture with a learning rate of 0.001 achieved the global minimum for validation loss.

Hidden Size	LR = 0.0005	LR = 0.001	LR = 0.01
32	0.007770	0.007284	0.007008
64	0.007307	0.007024	0.007112
128	0.007066	0.006723	0.007091

Figure 3: Hyperparameter grid search resulting in a LR of 0.001 and a hidden size of 128.

5 Baseline Model

The baseline is a standard Multiple Linear Regression model implemented as a single fully-connected neural network layer.

Instead of reading the data chronologically, the model uses `nn.Flatten()` to turn the 14-day

matrix into a single 168-element vector. It then maps those 168 (14 days $\times$ 12 features) inputs directly to the 7 future output days.

It establishes the absolute minimum performance threshold to prove whether a deep learning architecture is worth the computational cost. The baseline suffers from a complete loss of temporal perception, seeing the dataset as a bag of features, making it highly sensitive to noisy data.

6 Quantitative Results

6.1 Baseline

The baseline was trained for 30 epochs using the Adam optimizer, reaching a validation MSE low of 0.0078 at epoch 15.

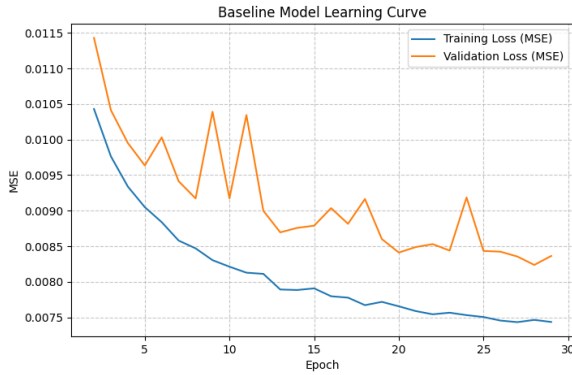


Figure 4: Linear Baseline Learning Curve. The validation loss (orange) remains highly unstable, indicating poor generalization to non-linear shifts in weather data.

Table 1: Baseline Training Progress (Selected Epochs)

Epoch	Train MSE	Val MSE
5	0.0091	0.0097
10	0.0083	0.0104
15	0.0079	0.0078
20	0.0076	0.0084
25	0.0075	0.0084
30	0.0074	0.0083

6.2 GRU Model

The finalized GRU model achieved a **Final Test MSE of 0.006066** on the out-of-sample 2022–2025 dataset. This indicates a high level of predictive accuracy and robust generalization to post-pandemic load shifts.

7 Qualitative Results

7.1 Baseline

As shown in Figure 6, the baseline successfully captures the general demand magnitude but fails to accurately replicate the specific "peaks" and "troughs" of the 7-day horizon. Because the linear model lacks recurrent memory, it is overly sensitive to local variance. A single day of outlier weather in a batch can cause a significant weight shift, leading to the erratic performance.

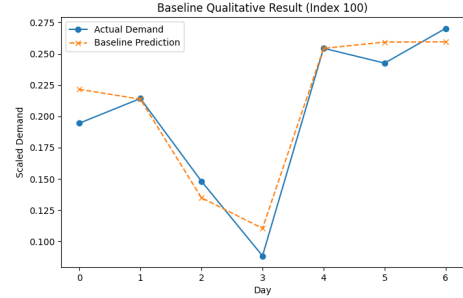


Figure 6: Baseline qualitative 7-day forecast (Index 100). The model underestimates the demand drop on Day 3 and the peak on Day 6. However, Day 4 was very close

7.2 GRU Model

Qualitatively, the model performance was evaluated using an out-of-sample window from the test set. Figure 7 displays the prediction for **Index 100**, corresponding to mid-April 2022. This period is particularly significant as it represents a seasonal transition in Ontario, characterized by volatile temperature fluctuations. The GRU successfully tracks the morphological "rhythm" of the demand curve; however, it exhibits a slight conservative bias, underpredicting the absolute magnitude of the mid-week peak and the depth of the weekend trough.

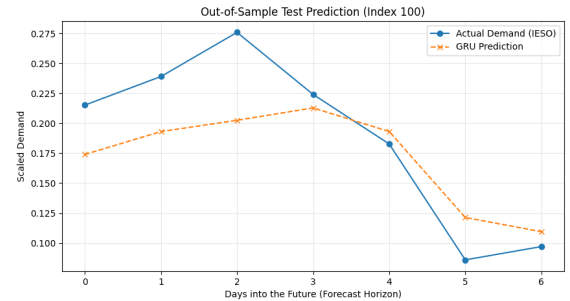


Figure 7: Out-of-Sample Test Prediction for Index 100 (April 2022).

Although the linear baseline provides a reasonable approximation of the weekly cycle, the GRU demonstrates superior morphological

tracking, evidenced by the 25% reduction in Test MSE.

8 Discussion

8.1 Model Evaluation on New Data

As discussed in class, models need to be trained on entirely unseen data to ensure proper learning and avoid memorization. The splitting strategy was explained in Section 3. The final evaluation was conducted on a localized test set from 2022 to 2025. This era occurred immediately following the height of the COVID-19 pandemic, a period where energy usage was highly irregular and lockdown policies drastically altered commercial electricity consumption [14]. Nevertheless, the final model achieved a Test MSE of 0.006066, proving the GRU did not merely memorize the training data, but successfully generalized to post-pandemic load shifts and compounding weather effects that it was never explicitly trained on.

8.2 Baseline Versus GRU

A fundamental limitation of the linear baseline lies in its spatial perception. By flattening the 14×12 tensor into a single vector of 168 features, the baseline treats the dataset as an unordered pile. Mathematically, it loses the sequential relationship between days.

In contrast, the primary GRU model views the data as a chronological journey. It processes the 14-day window step-by-step, utilizing internal "gates" to maintain a hidden memory state. This allows the model to acknowledge physics-based realities—such as heat accumulating in the power grid over several consecutive days—while successfully ignoring the one-day stochastic weather outliers that caused the extreme validation volatility observed in the baseline’s learning curve.

8.3 Implementation Challenges

A significant challenge involved balancing model sensitivity with stability. Initial tests with-

out dropout and with redundant features (16 variables) resulted in severe training instability. This was mitigated by removing unnecessary features, increasing the dropout to 0.4, and adding an additional GRU layer.

8.4 Ethical Considerations

The proposed system raises ethical considerations. Fine-grained electric load data can indirectly reveal patterns of human activity, raising potential privacy concerns if applied at the household level. Care must be taken to ensure data is anonymous and complies with relevant protection standards.

Underestimation in peak demand predictions could lead to insufficient grid preparation, increasing the risk of outages. Additionally, overestimation could result in unnecessary operational costs or inefficient resource allocation. Since the model’s performance depends on the representativeness of the training data, demand patterns learned from one region may not represent others well, especially in areas with rapidly changing infrastructure or energy access.

8.5 Project Difficulty

Firstly, the data pipeline was highly non-trivial, as it required merging three disparate, real-world datasets with mismatched temporal resolutions (hourly demand, hourly weather, and monthly population growth) into a unified daily format. Handling time-series data required generating sliding-window 3D PyTorch tensors.

Secondly, the objective of predicting a continuous 7-day future horizon is more complex than single-step forecasting. The model had to learn socio-economic weekly patterns while adapting to highly stochastic, non-linear weather compounding effects.

Lastly, attempting to solve the non-trivial "mean-seeking" bias of standard MSE loss by implementing Asymmetric Loss and Max-Pooling concatenations was quite difficult.

Despite all this, the finalized GRU successfully captured the rhythm of real-world energy grids, achieving a 25% reduction in error compared to the spatial baseline.

References

- [1] Y. He, J. Wang, et al. Electricity consumption peak demand prediction based on hybrid deep learning models. *Electrical Engineering*, 107:1–15, 2025. URL: <https://link.springer.com/article/10.1007/s00202-025-03437-5>, doi:10.1007/s00202-025-03437-5.
- [2] Angelina Yee. Deep learning for short-term electricity peak demand prediction, 2026. URL: <https://github.com/angelina-e-yee/Deep-Learning-for-Short-Term-Electricity-Peak-Demand-Prediction>.
- [3] Saeed Alqahtani, Imran Khan, and Saif Rehman. Advanced deep learning algorithms for energy optimization of smart cities. *Energies*, 18(2):407, 2025.
- [4] Xuefeng Wang, Yu Zhang, and Zhigang Li. Electric load forecasting based on deep learning and optimized by heuristic algorithm in smart grid. *Applied Energy*, 277:115495, 2020.
- [5] Mohd Abdullah, Mohd Jamil, and Abid Hussain. Forecasting household electric appliances consumption and peak demand based on hybrid machine learning approach. *Journal of Building Engineering*, 32:101496, 2020.
- [6] Muhammad Shahbaz, Muhammad Zeshan, and Amjad Javed. A hybrid data mining driven algorithm for long term electric peak load and energy demand forecasting. *Energy*, 203:117791, 2020.
- [7] Rami Al-Ruzouq, Abdallah Shanableh, Tariq Al-Mansoori, Ghaith Al-Khayyat, and Ali Al-Lami. Advanced machine learning techniques for energy optimization and load forecasting. *Energy Reports*, 11:5550–5568, 2024. doi:10.1016/j.egyr.2024.05.034.
- [8] Ghulam Hafeez, Muhammad Alimgeer, Ibrar Khan, Muhammad Usman, Fawad Khan, Zahid Wadud, and Bilal Khan. Electric load forecasting based on deep learning and optimized by heuristic algorithm in smart grid. *Applied Energy*, 269:114915, 2020. doi:10.1016/j.apenergy.2020.114915.
- [9] Jacob Sharples. Ontario electricity demand, 2021. Hourly provincial electricity demand data for Ontario in kWh. URL: <https://www.kaggle.com/datasets/jacobsharples/ontario-electricity-demand>.
- [10] Environment and Climate Change Canada. Historical climate data, 2024. Hourly meteorological observations for Ontario weather stations including Toronto. URL: https://climate.weather.gc.ca/historical_data/.
- [11] Ali Ranjipour. Toronto daily weather data (1937–2025), 2025. Daily weather data for Toronto. URL: <https://www.kaggle.com/datasets/aliranjipour/toronto-weather-stats-nov-1937-jul-2025>.
- [12] N. Goutam. Ontario birth and mortality data (1994–2024). <https://www.kaggle.com/datasets/goutamnr/ontario-birth-and-mortality-data-19942024>, 2024.
- [13] Arvind R. Singh, M. S. Sujatha, Akshay D. Kadu, Mohit Bajaj, Hailu Kendie Addis, and Kota Sarada. A deep learning and iot-driven framework for real-time adaptive resource allocation and grid optimization in smart energy systems. *Scientific Reports*, 15:19309, 2025. doi:10.1038/s41598-025-02649-w.
- [14] Farzam Kharvari a William O’Brien a Burak Gunay a Ahmed Abdeen a, b. The impact of the covid-19 on households’ hourly electricity consumption in canada. <https://pmc.ncbi.nlm.nih.gov/articles/PMC8797011/>.